

Class 08

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Class 8 Mini-Project

Preparing the data

```
# Save your input data file into your Project directory  
fna.data <- "WisconsinCancer.csv"
```

```
# Inputting the data and store as wisc.df  
wisc.df <- read.csv(fna.data, row.names=1)
```

```
#Examining input data to ensure column names are set correctly  
head(wisc.df, 4)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302	0.11840	0.27760	0.3001	0.14710	
842517	0.08474	0.07864	0.0869	0.07017	
84300903	0.10960	0.15990	0.1974	0.12790	
84348301	0.14250	0.28390	0.2414	0.10520	
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
84348301	0.2597	0.09744	0.4956	1.1560	3.445
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340

84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003	0.006193	25.38	17.33	
842517	0.01389	0.003532	24.99	23.41	
84300903	0.02250	0.004571	23.57	25.53	
84348301	0.05963	0.009208	14.91	26.50	
	perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302	184.60	2019.0	0.1622	0.6656	
842517	158.80	1956.0	0.1238	0.1866	
84300903	152.50	1709.0	0.1444	0.4245	
84348301	98.87	567.7	0.2098	0.8663	
	concavity_worst	concave.points_worst	symmetry_worst		
842302	0.7119	0.2654	0.4601		
842517	0.2416	0.1860	0.2750		
84300903	0.4504	0.2430	0.3613		
84348301	0.6869	0.2575	0.6638		
	fractal_dimension_worst				
842302	0.11890				
842517	0.08902				
84300903	0.08758				
84348301	0.17300				

```
#Using -1 to remove the first column
wisc.data <- wisc.df[,-1]
```

```
# Create diagnosis vector for later
diagnosis <- as.factor(wisc.df$diagnosis)
```

Exploratory data analysis

Q1. How many observations are in this dataset?

```
#Using nrow to show the number of rows in the dataset
nrow(wisc.df)
```

```
[1] 569
```

There are 569 observations in this data set

Q2. How many of the observations have a malignant diagnosis?

```
#Using table() to count how many times each category appears in diagnosis vector
table(diagnosis)
```

```
diagnosis
  B    M
357 212
```

There are 212 malignant diagnoses.

Q3. How many variables/features in the data are suffixed with `_mean`?

```
#Using grep() to find the column names that end with _mean.
#Using length() to find the amount of matches
grep("_mean$", colnames(wisc.df), value = TRUE)
```

```
[1] "radius_mean"          "texture_mean"         "perimeter_mean"
[4] "area_mean"            "smoothness_mean"      "compactness_mean"
[7] "concavity_mean"       "concave.points_mean"  "symmetry_mean"
[10] "fractal_dimension_mean"
```

```
length(grep("_mean$", colnames(wisc.df)))
```

```
[1] 10
```

10 of the variables/features in the data are suffixed with `_mean`.

3. Principal Component Analysis

Performing PCA

```
#Using -1 to remove the first column
wisc.data <- wisc.df[,-1]
# Checking column means and standard deviations
colMeans(wisc.data)
```

radius_mean	texture_mean	perimeter_mean
1.412729e+01	1.928965e+01	9.196903e+01
area_mean	smoothness_mean	compactness_mean
6.548891e+02	9.636028e-02	1.043410e-01
concavity_mean	concave.points_mean	symmetry_mean

8.879932e-02	4.891915e-02	1.811619e-01
fractal_dimension_mean	radius_se	texture_se
6.279761e-02	4.051721e-01	1.216853e+00
perimeter_se	area_se	smoothness_se
2.866059e+00	4.033708e+01	7.040979e-03
compactness_se	concavity_se	concave.points_se
2.547814e-02	3.189372e-02	1.179614e-02
symmetry_se	fractal_dimension_se	radius_worst
2.054230e-02	3.794904e-03	1.626919e+01
texture_worst	perimeter_worst	area_worst
2.567722e+01	1.072612e+02	8.805831e+02
smoothness_worst	compactness_worst	concavity_worst
1.323686e-01	2.542650e-01	2.721885e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
1.146062e-01	2.900756e-01	8.394582e-02

```
apply(wisc.data,2,sd)
```

radius_mean	texture_mean	perimeter_mean
3.524049e+00	4.301036e+00	2.429898e+01
area_mean	smoothness_mean	compactness_mean
3.519141e+02	1.406413e-02	5.281276e-02
concavity_mean	concave.points_mean	symmetry_mean
7.971981e-02	3.880284e-02	2.741428e-02
fractal_dimension_mean	radius_se	texture_se
7.060363e-03	2.773127e-01	5.516484e-01
perimeter_se	area_se	smoothness_se
2.021855e+00	4.549101e+01	3.002518e-03
compactness_se	concavity_se	concave.points_se
1.790818e-02	3.018606e-02	6.170285e-03
symmetry_se	fractal_dimension_se	radius_worst
8.266372e-03	2.646071e-03	4.833242e+00
texture_worst	perimeter_worst	area_worst
6.146258e+00	3.360254e+01	5.693570e+02
smoothness_worst	compactness_worst	concavity_worst
2.283243e-02	1.573365e-01	2.086243e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
6.573234e-02	6.186747e-02	1.806127e-02

```
#Perfoming PCA on wisc.data
wisc.pr <- prcomp(wisc.data, scale = TRUE)
```

```
#Looking at the PCA summary results
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

The proportion of the original variance captured by the PC1 is 0.4427, or 44.27%.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

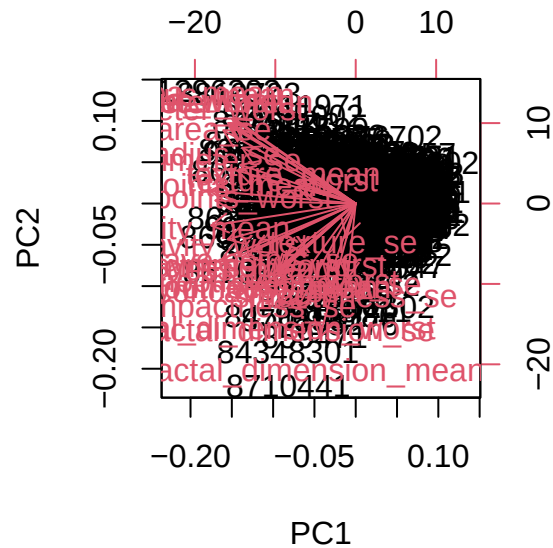
Looking at the cumulative proportion, PC3 is the first time the total variance explained exceeds 70%, since it reaches 0.72636 or ~72%, therefore only 3 principal components are required.

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

Looking at the cumulative proportion, PC7 is the first time the total variance explained exceeds 90%, since it reaches 0.91010 or ~91%, therefore only 7 principle components are required.

Interpreting PCA results

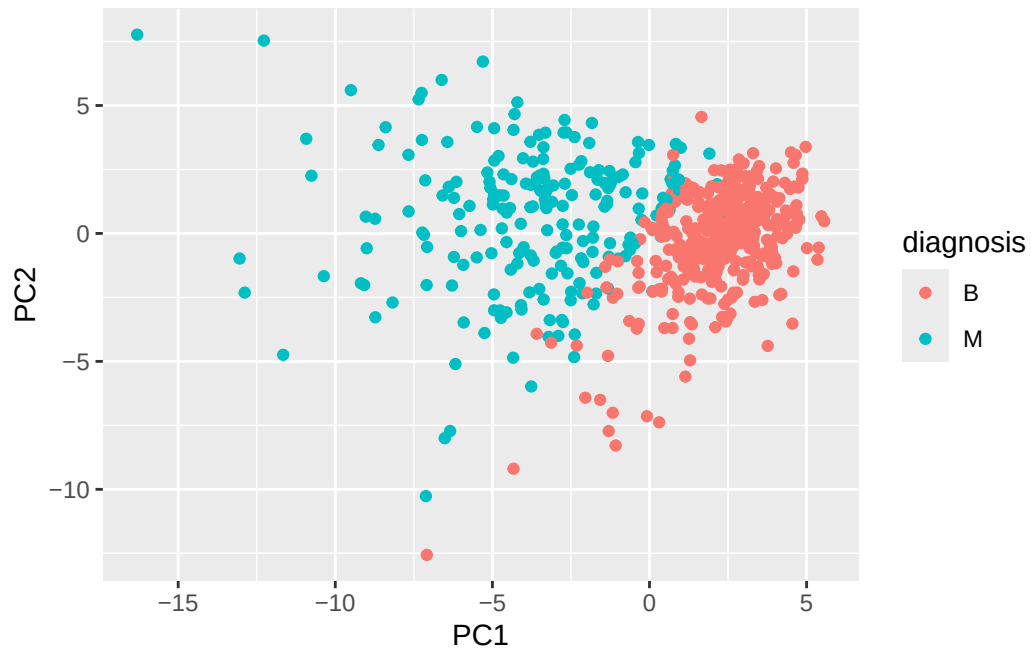
```
biplot(wisc.pr)
```



Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

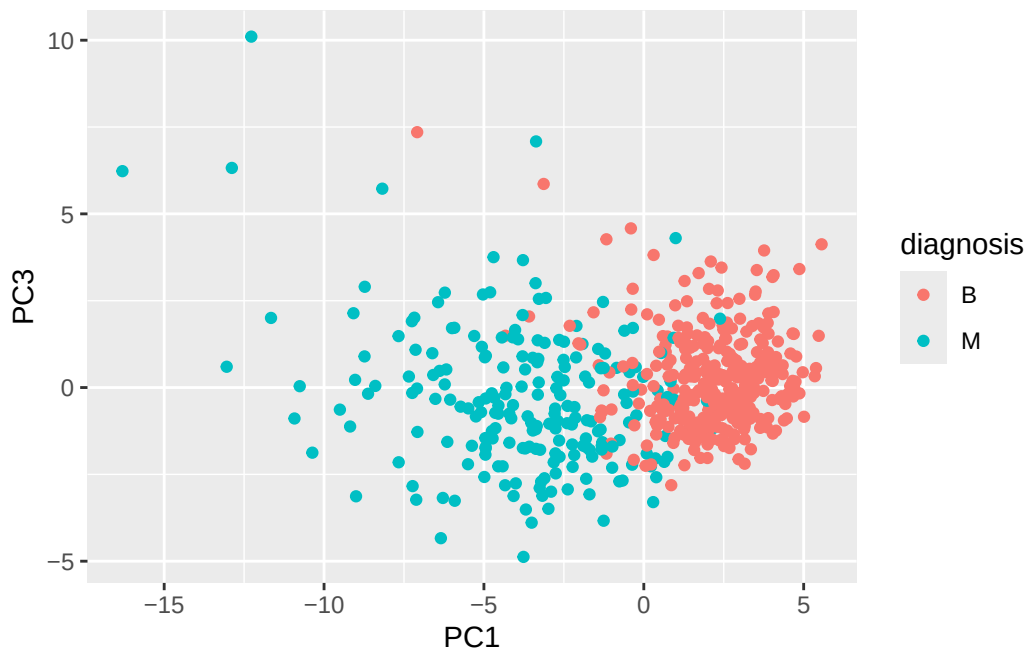
What stands out to me about this biplot is there are many overlapping numbers and labels. This makes it hard to understand.

```
#Loading in ggplot2
library(ggplot2)
#Generating a scatter plot of the first two principal components
ggplot(wisc.pr$x) +
  #Setting PC1 for x-axis and PC2 for y-axis and coloring points by diagnosis
  aes(PC1, PC2, col = diagnosis) +
  #Adding points for each observation
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
#Generating scatter plot of PC1 and PC3
ggplot(wisc.pr$x) +
#Setting P1 to x-axis and PC3 to y-axis and color points by diagnosis
  aes(PC1, PC3, col = diagnosis) +
  #Adding points for each observation
  geom_point()
```



From the plots I notice that PC1 is better at separating malignant from benign samples in both plots. However, the PC1 vs PC3 plot has more overlap in points than PC1 vs PC2, suggesting that PC3 is less effective at separating between the malignant and benign diagnoses.

Variance Explained

```
#Calculating the variance of each principal component
#by squaring the sdev component of wisc.pr
pr.var <- wisc.pr$sdev^2
```

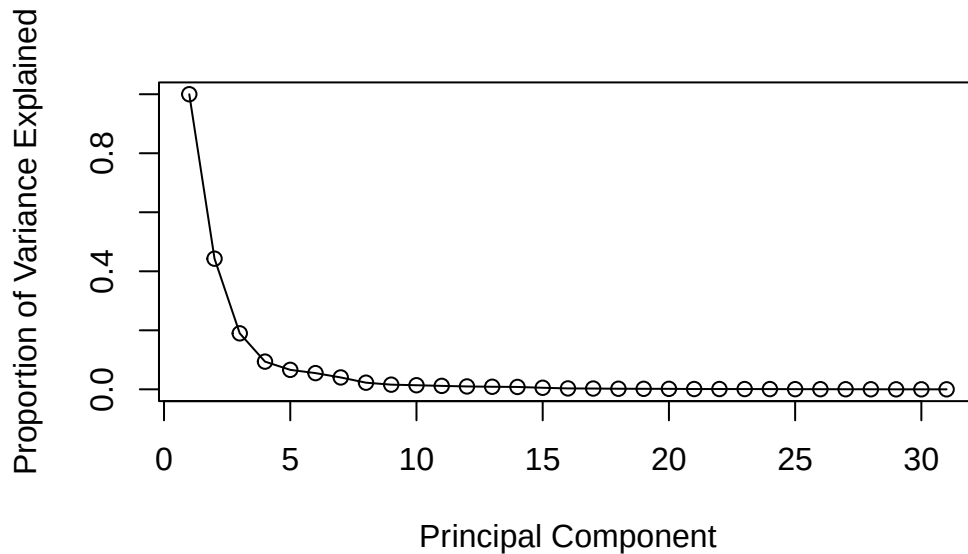
```
#Viewing the first few values
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

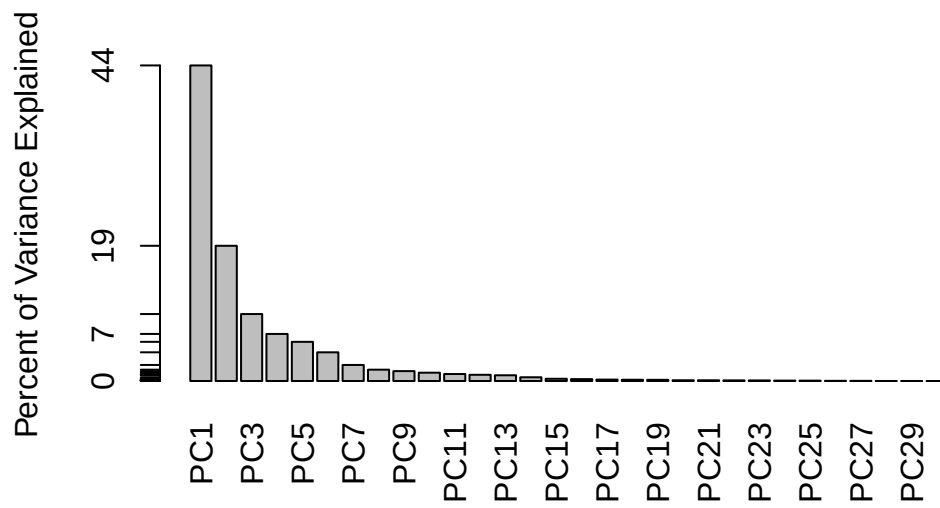
```
#Calculating the variance explained by each principal component
#Dividing each PC variance by the total variance to get proportion explained
pve <- pr.var / sum(pr.var)
#Plotting the variance explained for each principal component
plot(c(1,pve),
     #Labeling x-axis and y-axis
     xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
```



```
#Setting the y-axis limits from 0 to 1
ylim = c(0,1),
#Plotting both the points with a trend line
type = "o")
```



```
# Alternative scree plot of the same data, note data driven y-axis
#Generating a barplot
barplot(pve, ylab = "Percent of Variance Explained",
        #Labeling each bar as PC1, PC2, etc. using names.arg
        names.arg=paste0("PC",1:length(pve)),
        #Rotating x-axis labels so they are vertical and removing default y-axis
        las=2, axes = FALSE)
#Adding custom y-axis on left side
axis(2,
     # Puts marks on each PVE value
     at=pve,
     #Converting proportions to percentages and rounding them
     labels=round(pve,2)*100 )
```

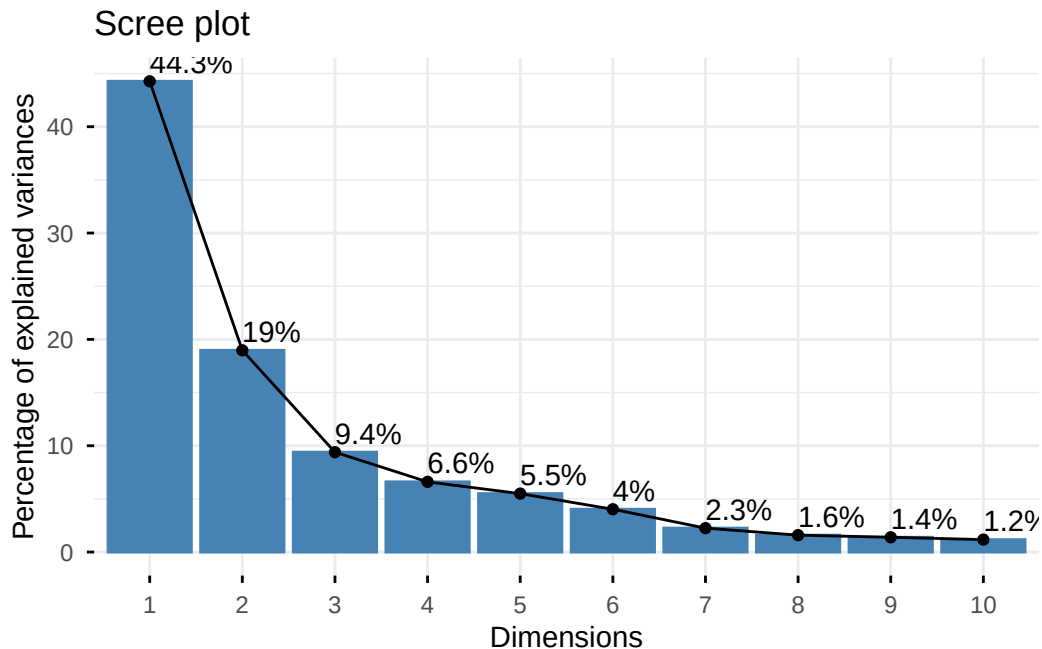


```
## ggplot based graph
#install.packages("factoextra")
library(factoextra)
```

Welcome to factoextra!

Want to learn more? See two factoextra-related books at <https://www.datanovia.com/en/product/guide-to-principal-component-methods-in-r/>

```
#Generating scree plot
fviz_eig(wisc.pr, addlabels = TRUE)
```



Communicating PCA results

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
#Retrieving the PC1 loading for the feature concave.points_mean
#to see its contribution to PC1
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

```
#Viewing all PC1 loadings and sort by size
sort(wisc.pr$rotation[,1], decreasing = TRUE)
```

smoothness_se	texture_se	symmetry_se
-0.01453145	-0.01742803	-0.04249842
fractal_dimension_mean	fractal_dimension_se	texture_mean
-0.06436335	-0.10256832	-0.10372458
texture_worst	symmetry_worst	smoothness_worst
-0.10446933	-0.12290456	-0.12795256

fractal_dimension_worst	symmetry_mean	smoothness_mean
-0.13178394	-0.13816696	-0.14258969
concavity_se	compactness_se	concave.points_se
-0.15358979	-0.17039345	-0.18341740
area_se	radius_se	compactness_worst
-0.20286964	-0.20597878	-0.21009588
perimeter_se	radius_mean	area_mean
-0.21132592	-0.21890244	-0.22099499
area_worst	perimeter_mean	radius_worst
-0.22487053	-0.22753729	-0.22799663
concavity_worst	perimeter_worst	compactness_mean
-0.22876753	-0.23663968	-0.23928535
concave.points_worst	concavity_mean	concave.points_mean
-0.25088597	-0.25840048	-0.26085376

On the PC1, the loading value for concave.points_mean is -0.2608538. This suggests that this feature contributes strongly to PC1. There are no features with larger contributions than this one.

4. Hierarchical Clustering

```
#Scaling the wisc.data
data.scaled <- scale(wisc.data)
```

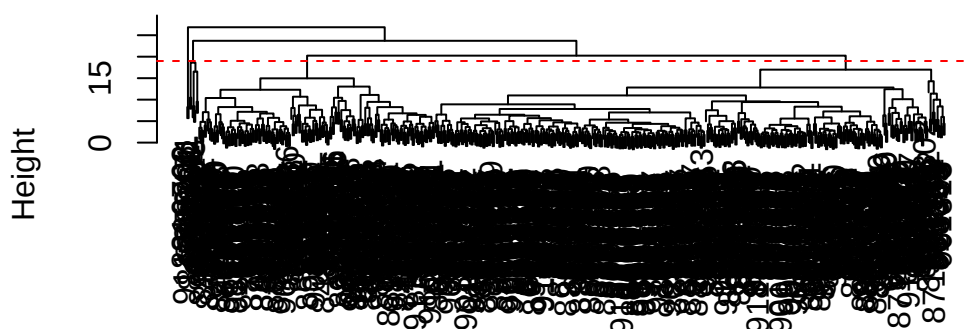
```
#Calculating Euclidean distances between observations
data.dist <- dist(data.scaled)
```

```
#Creating a hierarchical clustering model using complete linkage
wisc.hclust <- hclust(data.dist, method = "complete")
```

Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```
#Plotting the hierarchical clustering dendrogram
plot(wisc.hclust)
abline(h=19, col = "red", lty = 2)
```

Cluster Dendrogram



```
data.dist
hclust (*, "complete")
```

The height at which the clustering model has 4 clusters is around 19.

Selecting number of clusters

```
#Cutting the hierarchial clustering tree to 4 clusters
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)
#zGenerating table comparing cluster assignments to true labels
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

Q11. OPTIONAL: Can you find a better cluster vs diagnoses match by cutting into a different number of clusters between 2 and 6? How do you judge the quality of your result in each case?

```
#Trying k = 2 to 6 to comapre each diagnosis
for (k in 2:6) {
  clusters <- cutree(wisc.hclust, k = k)
```

```
print(table(clusters, diagnosis))
}
```

```
      diagnosis
clusters  B    M
1  357 210
2    0    2
```

```
      diagnosis
clusters  B    M
1  355 205
2    2    5
3    0    2
```

```
      diagnosis
clusters  B    M
1   12 165
2    2    5
3  343   40
4    0    2
```

```
      diagnosis
clusters  B    M
1   12 165
2    0    5
3  343   40
4    2    0
5    0    2
```

```
      diagnosis
clusters  B    M
1   12 165
2    0    5
3  331   39
4    2    0
5   12    1
6    0    2
```

The best match occurred at $k = 5$, where the clusters are more pure with less mixing than the other clusters. The quality of clustering was measured by comparing the clusters to diagnosis labels. The other clustering such as $k = 2$ and $k = 3$ did not have as good separation because these clusters are highly mixed, while $k = 6$ is split into many clusters without improving the separation between the diagnoses.

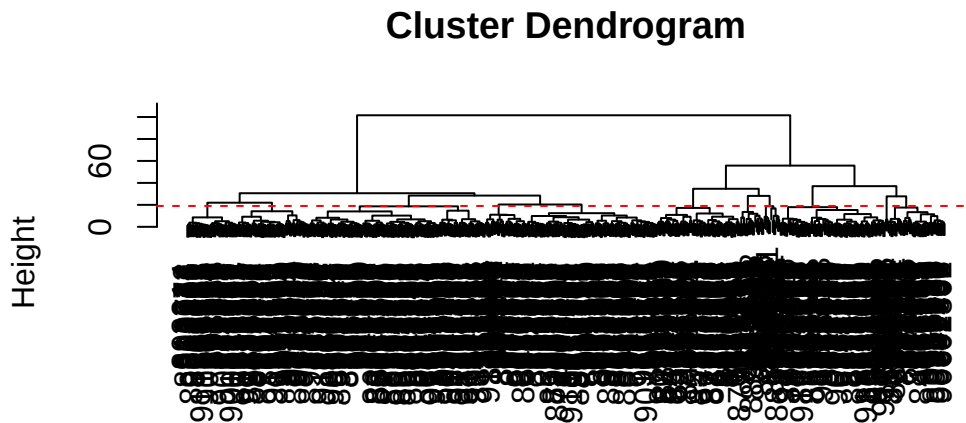
Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

The method that gives me my favorite results for the same data.dist dataset is ward.D2 because it makes the clusters more separated and organized without mixing the diagnoses of benign and malignant samples. This makes the clusters easier to interpret, and makes more logical sense for this biological scenario compared to the single, complete, or average methods.

5. Combining Methods

Comparing on PCA Results

```
#Using PC scores from wisc.pr$x to keep the minimum PCs needed for 90% variability
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")
#Plotting the hierarchial clustering dendrogram for wisc.pr.clust
plot(wisc.pr.hclust)
abline(h=19, col = "red", lty = 2)
```



```
dist(wisc.pr$x[, 1:7])
hclust (*, "ward.D2")
```

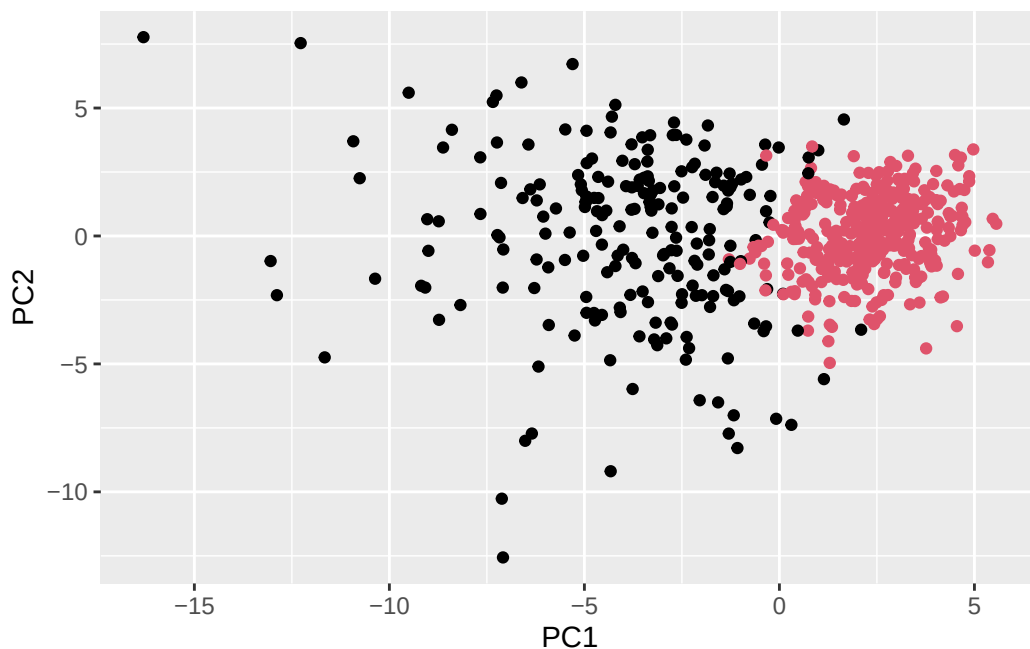
```
#Cutting the dendrogram to 2 clusters
grps <- cutree (wisc.pr.hclust, k = 2)
#Counting how many observatings are in each cluster
table(grps)
```

```
grps
  1  2
216 353
```

```
#Comparing the clusters to diagnoses
table(grps, diagnosis)
```

```
      diagnosis
grps   B     M
1    28 188
2   329  24
```

```
#Plotting PC1 vs PC2 and color each by its cluster assignment
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point (col = grps)
```



(Skipping the 3D part for pdf report)

```
#Using the distance along the first 7 principal components
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")
#Cutting PCA-based clustering into 2 clusters
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k = 2)
```

Q13. How well does the newly created hclust model with two clusters separate out the two “M” and “B” diagnoses?


```
#Comparing PCA-based clustering to actual diagnosis
table(wisc.pr.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.pr.hclust.clusters  B   M
1      28 188
2     329  24

```

This hclust model with two clusters separate out the two diagnoses well since one cluster is mostly malignant (188 M vs 28 B) while the other is mostly benign (329 B Vs 24 M). This demonstrates strong alignment with the actual diagnosis.

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```
#Comparing clustering without PCA
table(wisc.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.hclust.clusters  B   M
1      12 165
2       2   5
3     343  40
4       0   2

```

```
#Comparing clustering with PCA
table(wisc.pr.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.pr.hclust.clusters  B   M
1      28 188
2     329  24

```

The PCA based clustering model separates the malignant and benign samples more effectively than the non PCA based model. The PCA based clustering model shows clearer clusters with less overlap between the two diagnoses.

6. Sensitivity/Specificity

Q15. OPTIONAL: Which of your analysis procedures resulted in a clustering model with the best specificity? How about sensitivity?

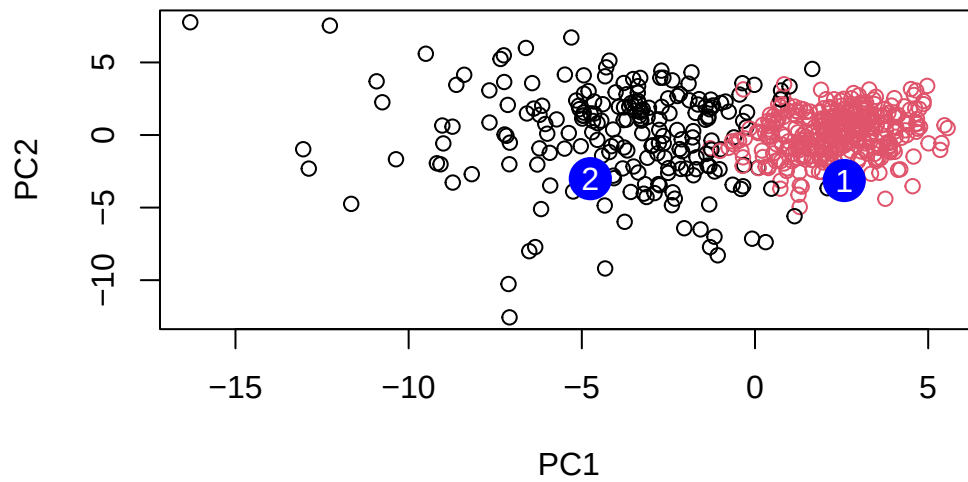
The PCA based clustering model had the best specificity with 92% and the best sensitivity with around 89%. This demonstrates that the model can effectively identify benign and malignant samples.

7. Prediction

```
#Inputting url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
#Plotting PC1 vs PC2 from the original data and coloring points by cluster assignment
plot(wisc.pr$x[,1:2], col=grps)
#Overlaying new patient samples in blue and label them 1 and 2
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Based on these results, the new patient we should prioritize is patient 1 since they are in the cluster associated with malignant samples. Patient 2 is in the benign cluster, so this case has less urgency.